



PRAKTIKUM

Pemrograman Dasar

Praktikum Kelas A

Asisten Praktikum

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Universitas Brawijaya**

2026

BAB 1

INPUT, PROSES, OUTPUT

1.6 Data dan Analisis Hasil Percobaan

1.6.1 Output

1. Jalankan Kelas Output1-Output4 dan benahi jika menemukan kesalahan!



The image consists of two screenshots of an IDE, likely IntelliJ IDEA, showing Java code and terminal output. The background of the IDE is a dark theme with a manga-style illustration.

Top Screenshot:

- File Explorer: `input.java 2` and `Output.java 1` are open.
- Code Editor: `Output.java` is selected. The code is:

```
3 public class Output {
4     public static void main(String[] args) {
5         System.out.println(x: "Hello World");
6     }
7 }
8 //Hashbi Abdillah
9 //265150600111022
```
- Terminal: Shows the command `java Output.java` and the output `Hello World`.

Bottom Screenshot:

- File Explorer: `prak2.java 2`, `variable.java 2`, `input.java 2`, and `Output2.java` are open.
- Code Editor: `Output2.java` is selected. The code is:

```
1 public class Output2 {
2     public static void main(String[] args) {
3         System.out.println(x: x: "Belajar");
4         System.out.println("pemrograman " + "Java");
5     }
6 }
7 //Hashbi Abdillah
8 //265150600111022
```
- Terminal: Shows the command `java Output2` and the output `Belajar` and `pemrograman Java`. There are yellow warning icons on the right side of the terminal output.

```
Output3.java > Output3 > main(String[])
1 public class Output3 {
2     public static void main(String[] args) {
3         System.out.printf(format: format: "%s%c\n", ...args: ...args: "Pemrograman
4         java", '|');
5         System.out.printf(format: format: "%30s%c\n", ...args: ...args: "Pemrograman
6         java", '|');
7         System.out.printf(format: format: "%-30s%c\n", ...args: ...args: "Pemrograman
8         java", '|');
9         System.out.printf(format: format: "%30.5s%c\n", ...args: ...args: "Pemrograman
10        java", '|');
11    }
12 }
13 //Hashbi Abdillah
14 //265150600111022

Problems 9 Output Debug Console Terminal Ports
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1> java Output3.java
Pemrograman java|
Pemrograman java|
Pemrograman java|
Pemrograman java|
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1>

Output4.java > Output4 > main(String[])
1 public class Output4 {
2     public static void main(String[] args) {
3         float x = 7654.123456789f;
4         System.out.printf(format: format: "%d %d %8d\n", ...args: ...args: 1234, -567, 8910)
5         System.out.printf(format: format: "%d %d %8d\n", ...args: ...args: 1234, -567, 8910)
6         System.out.printf(format: format: "%f %15f %15.3f\n", x, x, x);
7     }
8 }
9 //Hashbi Abdillah
10 //265150600111022

Problems 9 Output Debug Console Terminal Ports
7654,12 7,654124e+03 7654,124
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1> java Output4.java
1234 -567 8910
1234 -567 +8910
7654,123535 7654,123535 7654,124
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1>
```

2. error, karena terdapat 2 string tapi tidak digabungkan
- 3.a. line 3 mencetak line biasa secara normal, %s mencetak string dan %c mencetak karakter.
- 3.b. line 4 mencetak line rata kanan dengan alokasi 30 karakter karena menggunakan %30s dan %c mencetak karakter untuk penutup
- 3.c. line 5 adalah line 4, dibalik menjadi line rata kiri karena menggunakan %-30s
- 3.d. memotong hanya 5 karakter pertama karena ditambahkan .5, dicetak rata kanan

4. error karena jika menggunakan kutip 1 bukan bertipe string melainkan char yang dimana char hanya bernilai tidak boleh lebih dari 1 karakter

5. mencetak 8 karakter awal pada string dengan 30 space karakter, dengan format rata kiri dengan '|' sebagai penutup di sebelah kanan

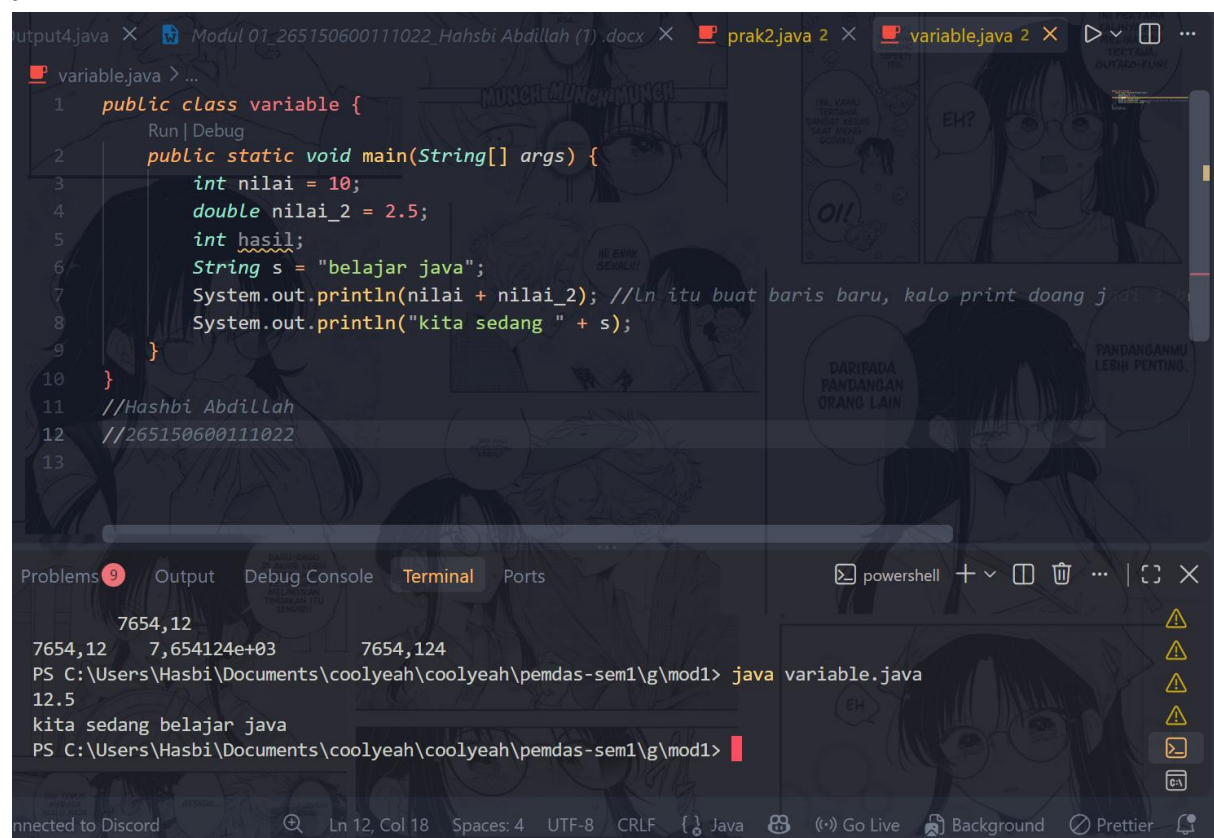
6. `System.out.printf("%15.2f\n", x);`

7. Kode tersebut mencetak angka dengan tiga format berbeda: %g (ringkas), %15e (eksponensial), dan %15.3f (desimal).

8. Kode ini menunjukkan cara mengatur tampilan angka saat dicetak, %d: menampilkan angka bulat, %f: menampilkan angka pecahan (desimal), %g: menampilkan angka secara ringkas, %e: menampilkan angka dalam bentuk eksponen (pangkat), %3d / %15f: mengatur lebar spasi angka, %.3f: membatasi berapa angka di belakang koma.

1.6.2 Variabel

9.



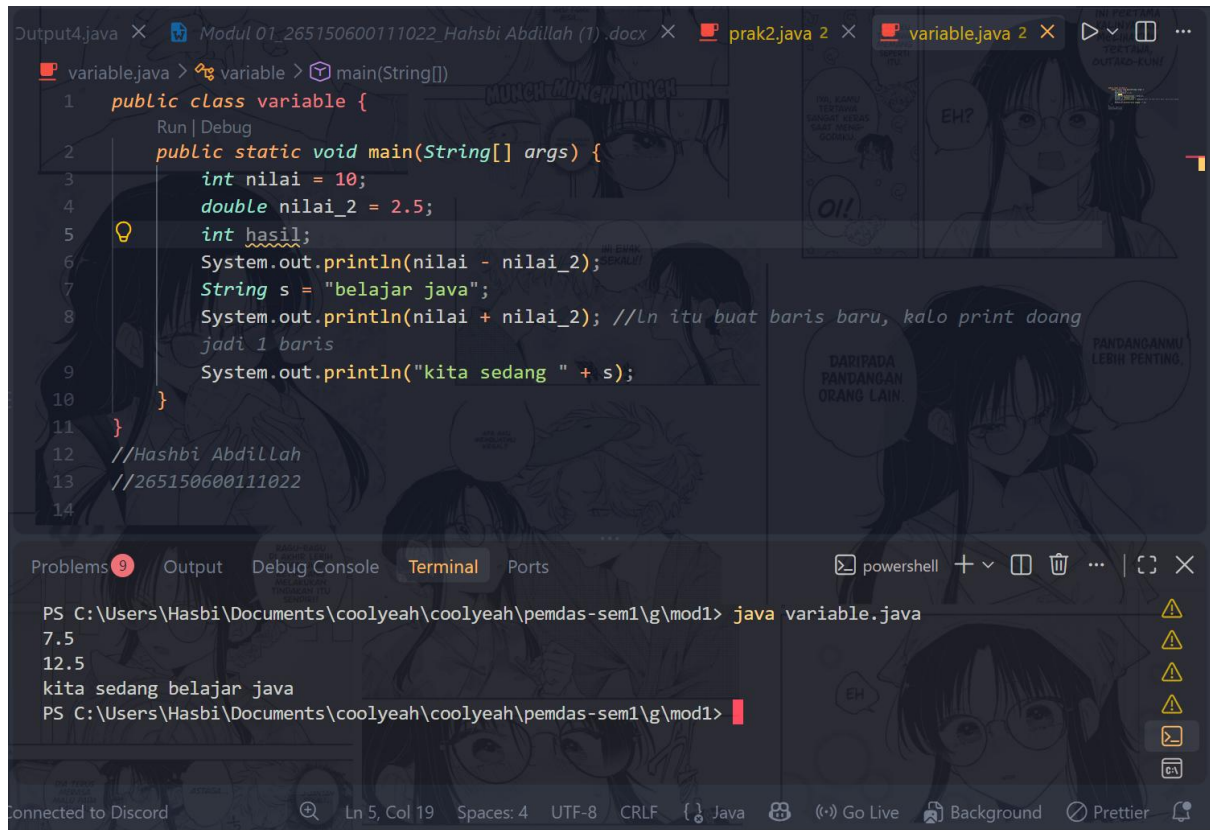
The screenshot shows an IDE with a Java file named `variable.java` and its output in the terminal. The code defines a class `variable` with a `main` method. It declares variables `nilai` (int, 10), `nilai_2` (double, 2.5), and `hasil` (int). It prints the sum of `nilai` and `nilai_2` using `println`, and then prints a string `s` using `println`. The terminal output shows the results of these operations.

```
1 public class variable {
2     public static void main(String[] args) {
3         int nilai = 10;
4         double nilai_2 = 2.5;
5         int hasil;
6         String s = "belajar java";
7         System.out.println(nilai + nilai_2); //ln itu buat baris baru, kalo print doang j...
8         System.out.println("kita sedang " + s);
9     }
10 }
11 //Hashbi Abdillah
12 //265150600111022
13
```

Terminal Output:

```
7654,12
7654,12 7,654124e+03 7654,124
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1> java variable.java
12.5
kita sedang belajar java
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>
```

10.



The screenshot shows an IDE with a Java file named `variable.java`. The code defines a `main` method that calculates the difference between two numbers, prints it, and then concatenates it with a string. The terminal output shows the results of these operations.

```
variable.java > variable > main(String[])
1 public class variable {
2     Run | Debug
3     public static void main(String[] args) {
4         int nilai = 10;
5         double nilai_2 = 2.5;
6         int hasil;
7         System.out.println(nilai - nilai_2);
8         String s = "belajar java";
9         System.out.println(nilai + nilai_2); //Ln itu buat baris baru, kalo print doang
10        jadi 1 baris
11        System.out.println("kita sedang " + s);
12    }
13    //Hashbi Abdillah
14    //265150600111022
```

Problems 9 Output Debug Console Terminal Ports

```
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-sem1\g\mod1> java variable.java
7.5
12.5
kita sedang belajar java
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-sem1\g\mod1>
```

Connected to Discord Ln 5, Col 19 Spaces: 4 UTF-8 CRLF Java Go Live Background Prettier

11. error karena char hanya bisa menyimpan 1 nilai karakter

```
Output4.java × Modul 01_265150600111022_Hahsbi Abdillah (1).docx × prak2.java 2 × variable.java × ! ▶ ◻ ...
variable.java > variable > main(String[])
1 public class variable {
2     public static void main(String[] args) {
3
4         double nilai_2 = 2.5;
5         int nilai_3 = 100;
6         int hasil = nilai_3 / 2;
7         System.out.println(nilai - nilai_2);
8         String s = "belajar java";
9         System.out.println(nilai + nilai_2); //ln itu buat baris baru, kalo print doang
10                                             jadi 1 baris
11         System.out.println("kita sedang " + s);
12         System.out.println(hasil);
13         // char ch = 'belajar java';
14     }
15     //Hashbi Abdillah
16     //265150600111022
}
```

Problems 7 Output Debug Console Terminal Ports powershell + ◻ ◻ ... | ◻ ×

```
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1> java variable.java
7.5
12.5
kita sedang belajar java
50
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>
```

Connected to Discord Ln 11, Col 33 Spaces: 4 UTF-8 CRLF { } Java Go Live Background Prettier

A screenshot of an IDE window. The top pane shows a Java file named 'variable.java' with the following code:

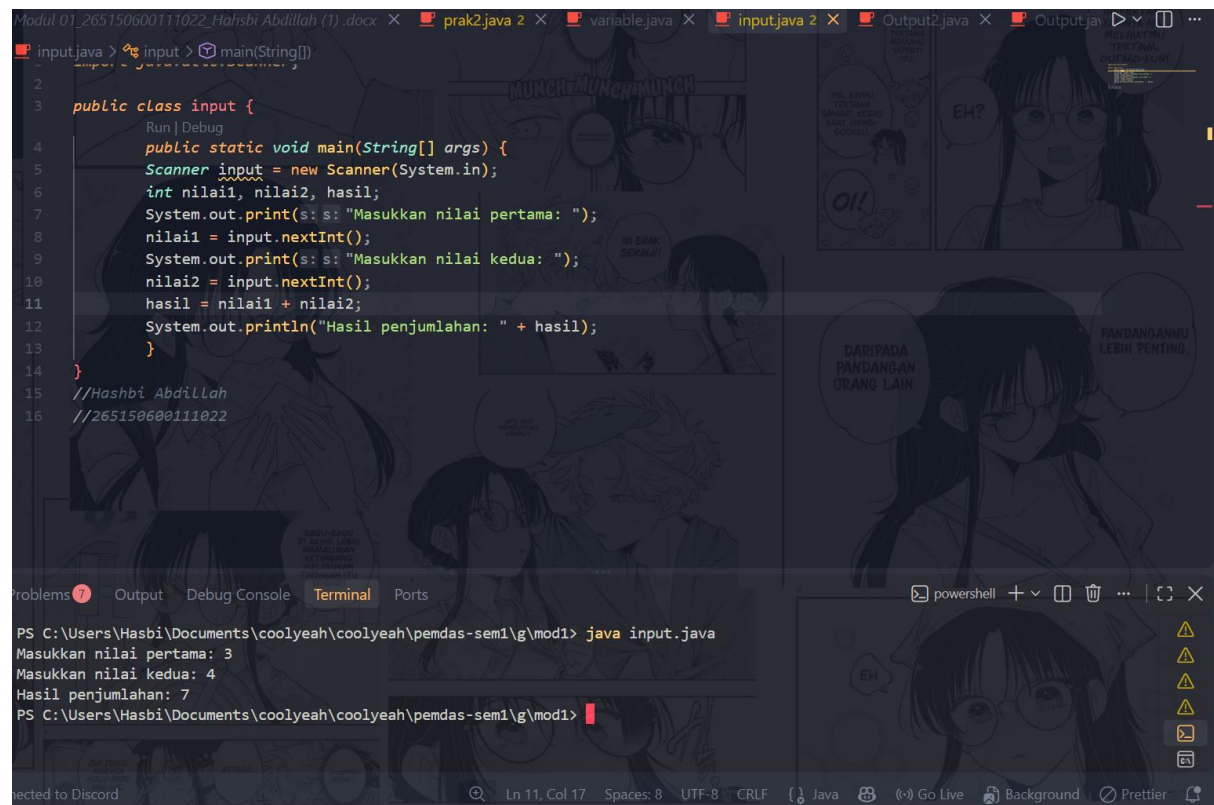
```
1 public class variable {  
2     public static void main(String[] args) {  
3         int nilai_3 = 100;  
4         int hasil = nilai_3 / 2;  
5         System.out.println(nilai - nilai_2);  
6         String s = "belajar java";  
7         System.out.println(nilai + nilai_2);  
8         System.out.println("kita sedang " + s);  
9         System.out.println(hasil);  
10        System.out.println(nilai_3 / 2);  
11        // char ch = 'belajar java';  
12    }  
13 }  
14  
15 //Hashbi Abdillah  
16 //265150600111022
```

The bottom pane is split into two sections. The left section, titled 'Problems', shows a list of errors: '7.5', '12.5', 'kita sedang belajar java', '50', '50', and 'PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pmdas-sem1\g\mod1>'. The right section, titled 'Terminal', shows the command prompt output: 'powershell', '7.5', '12.5', 'kita sedang belajar java', '50', '50', and 'PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pmdas-sem1\g\mod1>'. The IDE interface includes a top bar with file names, a left sidebar with a project tree, and a bottom status bar showing 'Connected to Discord', 'Ln 12, Col 36', 'Spaces: 4', 'UTF-8', 'CRLF', and 'Java'.

14. no 12 proses perhitungan dilakukan dengan memasukkan nilai ke variabel hasil, sedangkan no 13 perhitungan dilakukan sebelum mencetak hasil print tanpa menggunakan variabel hasil

1.6.3 Input

15.



```
Modul 01_265150600111022_Hahsbi Abdillah (1) .docx X prak2.java 2 X variable.java X input.java 2 X Output2.java X Outputja X ...
input.java > Run | Debug
main(String[])
2
3 public class input {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6         int nilai1, nilai2, hasil;
7         System.out.print(s: s: "Masukkan nilai pertama: ");
8         nilai1 = input.nextInt();
9         System.out.print(s: s: "Masukkan nilai kedua: ");
10        nilai2 = input.nextInt();
11        hasil = nilai1 + nilai2;
12        System.out.println("Hasil penjumlahan: " + hasil);
13    }
14 }
15 //Hashbi Abdillah
16 //265150600111022

Problems 7 Output Debug Console Terminal Ports
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1> java input.java
Masukkan nilai pertama: 3
Masukkan nilai kedua: 4
Hasil penjumlahan: 7
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>
```

16. Error karena tipe data yang di declare berupa integer

17.



The screenshot shows an IDE with a Java file named `input.java`. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class input {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6         int nilai1, nilai2, hasil;
7         String nama, nim;
8         System.out.print(s: s: "Masukkan nilai pertama: ");
9         nilai1 = input.nextInt();
10        System.out.print(s: s: "Masukkan nilai kedua: ");
11        nilai2 = input.nextInt();
12        input.nextLine();
13        System.out.print(s: s: "Masukkan nama: ");
14        nama = input.nextLine();
15        System.out.print(s: s: "Masukkan nim: ");
16        nim = input.nextLine();
17        hasil = nilai1 + nilai2;
18        System.out.println("Hasil penjumlahan: " + hasil);
19        System.out.println("nama: " + nama);
20        System.out.println("nim: " + nim);
21    }
22 }
```

The IDE's output window shows the execution results:

```
Hasil penjumlahan: 11
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-1\mod1> java input.java
Masukkan nilai pertama: 5
Masukkan nilai kedua: 4
Masukkan nama: hashbi abdillah
Masukkan nim: 265150600111022
Hasil penjumlahan: 9
nama: hashbi abdillah
nim: 265150600111022
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-1\mod1>
```

1.7 Tugas Praktikum


```
prak1.java X prak2.java X settings.java X
prak2.java 2 prak2 > @main(String[])
public class prak2 {
    public static void main(String[] args) {
        nama = input.nextLine();
        System.out.print(s: "Masukkan kelurahan: ");
        kelurahan = input.nextLine();
        System.out.print(s: "Masukkan Kwh Awal: ");
        kwhAwal = input.nextInt();
        System.out.print(s: "Masukkan Kwh Akhir: ");
        kwhAkhir = input.nextInt();
        System.out.print(s: "Masukkan biaya beban: ");
        biayaBeban = input.nextInt();
        System.out.print(s: "Masukkan ppj: ");
        ppj = input.nextInt();
        // System.out.printf("%sPLN %s%s\n", kelurahan);

        totPemakaian = kwhAkhir - kwhAwal;
        tarif = totPemakaian * biayaBeban;
        tarifppj = tarif * ppj / 100;
        totTarif = tarif + tarifppj;

        String textHeader = "PLN" + kelurahan, header = garisSisi + textHeader + garisSisi, footer = "-" + repeat(header.length());

        System.out.printf(format: "%s\n", header);
        System.out.println("Nama: " + nama);
        System.out.println("Kelurahan: " + kelurahan);
        System.out.println("Total pemakaian bulan ini: " + totPemakaian);
        System.out.println("Tarif listrik: " + tarif);
        System.out.printf("PPJ %d%%: " + tarifppj + "%n", ppj);
        System.out.println("Total bayar: " + totTarif);

        System.out.printf(format: "%s\n", footer);
    }
}

Problems Output Debug Console Terminal Ports
Masukkan biaya beban: 140
Masukkan ppj: 10
=====PLN Jawa utara=====
Nama: bakwan jagung
Kelurahan: Jawa utara
Total pemakaian bulan ini: 1000
Tarif listrik: 140000
PPJ 10%: 14000
Total bayar: 154000
=====
PS C:\Users\Hasbi\Documents\cool\yeah\cool\yeah\pemas-sen1\g\mod1>
```